Air Before and After Airing

Pupil sheet

Name: _____ **Date:** _____

Safety: Only the teacher opens the window; do not cover the particle sensor and do not use sprays near the box.

1. What I see on the screen

Our box has these sensors (tick): ☐ CO_2 ☐ particles (PM2.5, PM10) ☐ vapours (TVOC or the VOC index)

Three rules. I stay **at least a metre** from the box. I put nothing **above the particle sensor**. I press **Capture** once at the end of each phase.

2. What I do on my own

First I predict. I do not change my prediction after I measure.
When we open the window, these will fall: ☐ CO_2 ☐ the particles ☐ the vapours.

One of them might rise, because _____

_____

The three phases. I fill in only the rows for my box's sensors.

Quantity	Aired room	Full room	Window open
CO_2 (ppm)			
PM2.5 ($\mu g/m^3$)			
PM10 ($\mu g/m^3$)			
TVOC (ppb)			
VOC index			

How many of us were in the room: _____ Outside there was: ☐ a busy street ☐ a yard or park ☐ fog or smoke ☐ something else _____

The WHO guideline for PM2.5 is 15 $\mu g/m^3$ as a mean over a whole day. My value with the window open was: ☐ below ☐ above.

3. The conclusion I write

I predicted that, and the data show that _____

_____

The quantity that did not behave like the others was _____

_____

I think this happened because _____

_____

The limit of our measurement: why can we not compare ten minutes with the WHO's 24-hour guideline?

_____

Exit ticket

If there is fog and traffic outside, do you open the window or not? Why?

_____